

Saphenous Vein Autograft Replacement of Severe Segmental Coronary Artery Occlusion

Operative Technique

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Direct operation on the coronary artery has been performed in 180 patients at the Cleveland Clinic up to October, 1967. Recently, a new operative technique has been applied in 15 patients with extensive and severe obstruction of the right dominant coronary artery, specifically to overcome some of the unfavorable results that occurred when pericardial patch reconstruction was performed.

OPERATIVE TECHNIQUE

The operation is performed while the patient's circulation is maintained by total cardiopulmonary bypass at normal temperature. The left femoral artery is utilized for insertion of the arterial line, except in those patients with severe peripheral arteriosclerosis and in these the ascending aorta is cannulated. At the time the left femoral artery is exposed, using the same incision, the left saphenous vein is dissected free at the point where it joins the femoral vein. A segment of the saphenous vein is resected; a ligature is applied to the proximal end of the isolated segment, as an indicator of the direction of the flow, whereas the distal stump is left open. With this identification, it is easy to place the graft in the appropriate position (inverted), thus preventing impairment of the flow by the valves within the vein. The venous return is via individual cannulation of the superior and inferior venae cavae. A vent is obtained by inserting a catheter into the left atrium through the right superior pulmonary vein.

With the heart beating and the aorta unclamped, the right coronary artery is dissected, and sutures of 2-0 silk are placed above and below the obstruction, held by modified "baby gallbladder" clamps, the curves of which easily surround the vessel. After the artery is exposed, bulldog clamps (with rubber covers, to prevent tissue damage) are applied distal and proximal to the obstruction.

The uppermost surface of the obstruction is incised longitudinally until the lumen is reached. The proximal bulldog clamp can be released intermittently in order to facilitate the procedure. As soon as the lumen is entered, blood flow is evident. With a long, sharp, right-angled scissors, the incision is elongated distally and proximally until a good patent lumen is obtained (Fig. 1). Dilators are gently

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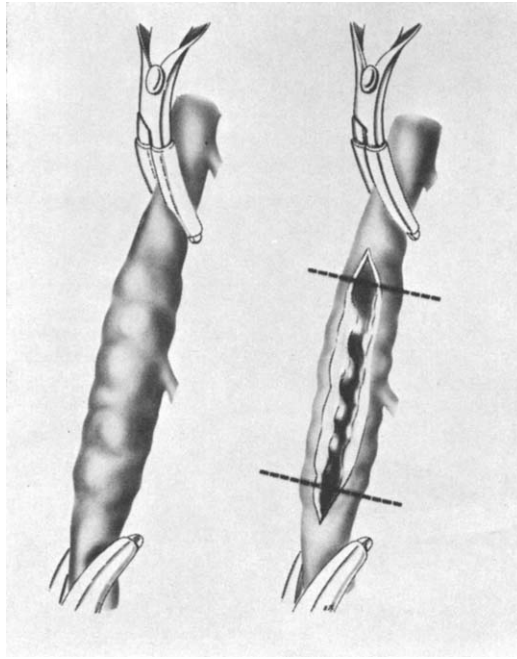


FIG. 1. On the left, the coronary artery is shown dissected and clamped. On the right side, a longitudinal incision is made on top of the obstruction and is elongated until a patent lumen is found. Dotted lines show the place of transection in order to obtain an enlarged stump.

introduced to ascertain whether or not further obstruction is present in either direction; this is the only reason for using a dilator. Enforced dilatation may tear the endothelium, and local thrombosis may ensue. The amount of backflow can be assessed by releasing the distal bulldog clamp. Significant backflow eliminates the need for distal perfusion; hence, the artery is clamped, and the surgical procedure continues without interruption. Inadequate return of blood necessitates perfusion of the distal segments through a small plastic cannula with blood at normal temperature.

The coronary artery is then transected about 2 or 3 mm. below the beginning of the longitudinal incision. This point of transection is an important landmark, as it is possible from there to enlarge the lumen of the stump and prevent narrowing of the coronary artery. The anastomosis is done with interrupted sutures of 6-0 silk. The first suture is placed posteriorly (Fig. 2); the saphenous vein is approximated, and the suture is tied. The rest of the sutures are placed alternately from one side to the other side of the first suture (Fig. 3). Approximately ten sutures are usually required; they should be placed close together to prevent leakage.

After the anastomosis is completed, the proximal bulldog clamp is released and the patency of the anastomosis is ascertained. Seldom must extra sutures be used when suturing has been meticulous. The bulldog clamp is next applied on the saphenous vein, and the distal portion of the stump is prepared in the same fashion. If distal perfusion is being used, it is discontinued at this moment. The distal anastomosis is performed in the same manner. The last three sutures are placed without tying them, to be sure that they are properly located. At this time the distal bulldog clamp is also released, and then two of the sutures are tied.

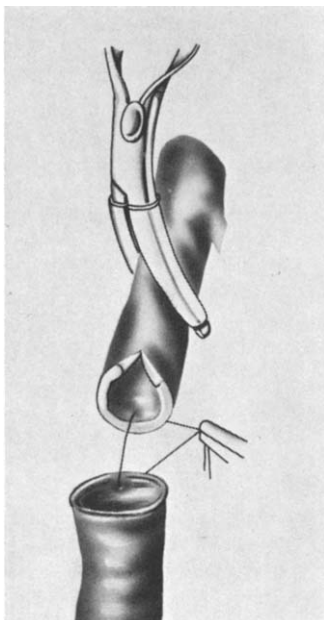


FIG. 2. The first suture is being placed posteriorly; the saphenous vein is approximated and the suture is tied.

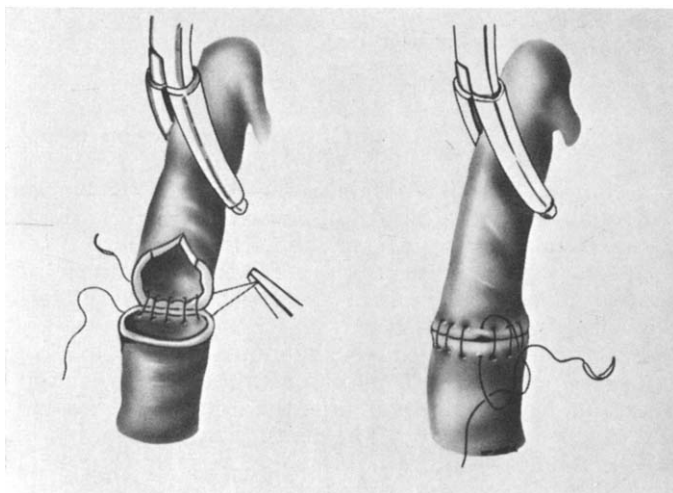


FIG. 3. The proximal anastomosis. On the left side, the sutures have been placed alternately from one side to the other side of the first suture. On the right, the completed anastomosis.

Before the last suture is tied, the proximal bulldog clamp is released so as to fill the lumen with blood and exclude any air trapped inside the coronary arteries (Fig. 4).

The anastomoses are performed with the aorta partially clamped; the heart continues to beat during the entire procedure. The coronary circuit receives blood of normal temperature. When the motion of the heart interferes with

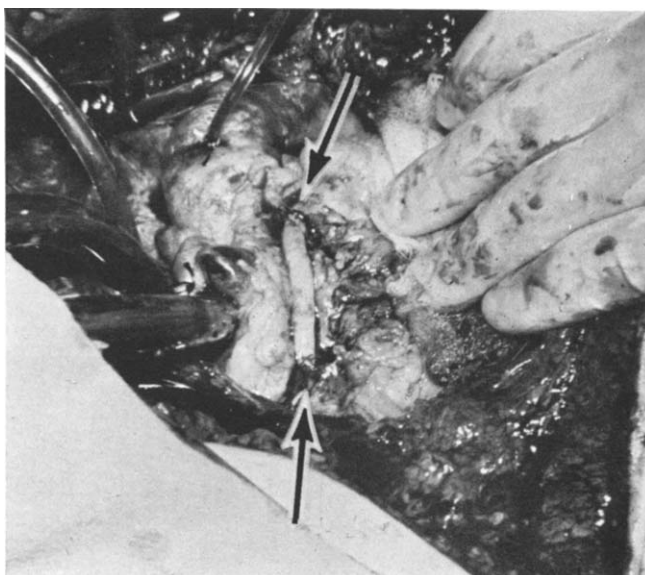


FIG. 4. Saphenous vein autograft at the end of the procedure.

proper placing of sutures, the aorta is totally clamped for no longer than 10 minutes, slowing the heart action and thus facilitating the surgical maneuvers.

RESULTS

This operation has been performed in 15 patients, all of whom are alive. Each of them had severe segmental occlusion of the right dominant coronary artery. One patient had previously undergone a right coronary endarterectomy. Three patients also benefited by simultaneously undergoing left internal mammary artery implantation, since severe obstruction of the left coronary artery system was present. The last patient had a 95% occlusion in the upper third of the right coronary artery, starting at the ostium. The saphenous vein graft was placed in the lower portion of the ascending aorta; there, a small opening was made and a side-to-end anastomosis was accomplished. The distal anastomosis was effected with the routine end-to-end anastomosis in the middle third of the right coronary artery. Six of these patients had undergone postoperative cine-coronary angiography studies showing excellent function of the graft (Figs. 5, 6).

COMMENT

A complete follow-up on the 180 patients who had had direct coronary artery surgery (coronary endarterectomy with pericardial patch graft) including selective coronary angiography (Sones' technique) showed that 13% exhibited significant narrowing, and 29% had total occlusion at the site of the patch. This occurred most often in patients with long segmental occlusions.

The indications for endarterectomy rather than endarterectomy have been discussed in previous reports [1-3], and obviously, at present,

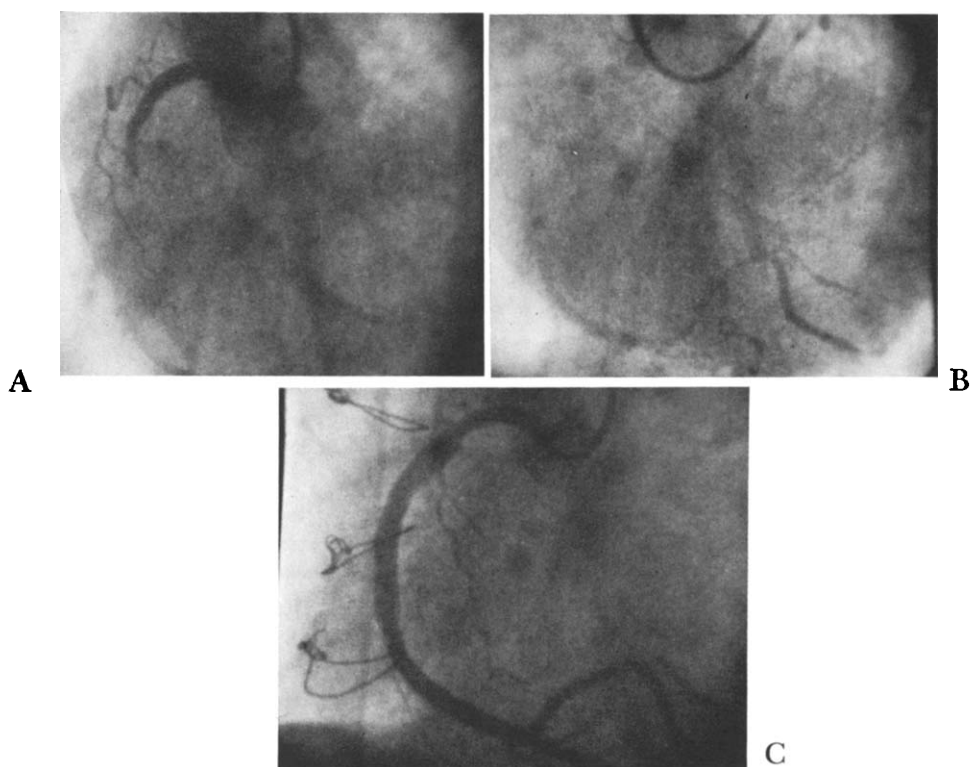


FIG. 5. *Coronary angiogram of a 55-year-old Caucasian woman with total occlusion of the right coronary artery (A). Perfusion of distal right coronary artery by collateral branches from the left coronary artery (B). Total reconstruction of right coronary artery by a saphenous vein autograft (C).*

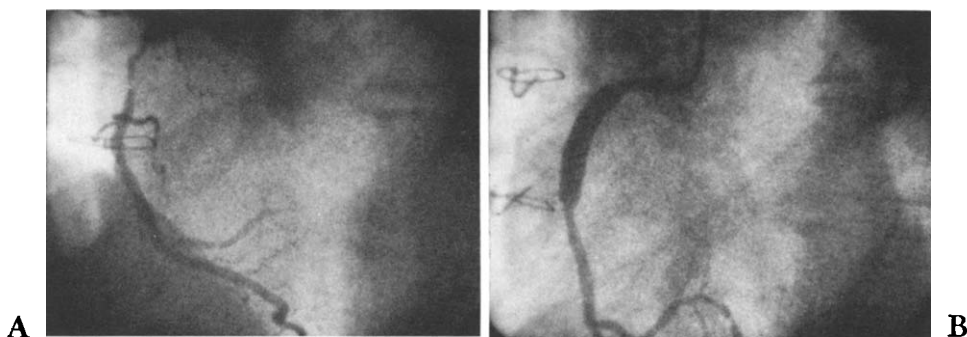


FIG. 6. *Coronary arteriogram of a 42-year-old Caucasian man who underwent a right coronary endarterectomy with pericardial patch graft in April, 1967. Post-operative catheterization shows severe segmental occlusion (A). A saphenous autograft replacement was performed in Sept., 1967. Catheterization on tenth post-operative day shows the vessel to be patent (B).*

true endarterectomy cannot be performed upon the coronary circulation. Placement of the pericardial patch graft on top of an irregular wall that usually has many crevices provides an ideal stratum for

further thromboses and occlusion. Saphenous vein interposition requires only two end-to-end anastomoses, thereby shortening the time of the operation, and the segment can be replaced by a smooth wall that will prevent occlusion. The concept that the coronary artery should be transected after a longitudinal incision is made is a fundamental and important step for two reasons: the transection can be done a few millimeters farther down, enlarging the lumen of the coronary artery stump; and crushing of the endothelium is prevented. In 1 patient in whom the transection was done before the artery was opened it was difficult to find the lumen. To facilitate placing of the sutures when the coronary artery is small, two sutures can be inserted in opposite sides, and by gentle traction the lumen can be kept open.

It is believed that performing the operation while the patient is being perfused with blood of normal temperature has significantly reduced the operative mortality and morbidity. We no longer use hypothermia and selective cooling of the heart on patients who undergo cardiac operations.

SUMMARY AND CONCLUSION

A new operative technique is described to correct severe segmental occlusion of the right dominant coronary artery. A saphenous vein autograft replaces the occluded arterial segment. Fifteen patients were operated upon without mortality. Postoperative angiographic catheterization has demonstrated excellent function of the grafts. Further application of the technique will help to determine the possibilities of its use on the left coronary artery in selected patients who have severe localized obstruction.

ADDENDUM

Since this paper was written, the total number of patients with whom this new operative technique has been applied has grown. The total number of cases is now 55. Fifty-two cases involved segmental occlusion of the right coronary artery; the other 3 involved the circumflex branch of the left coronary artery. Of the total, there were only 2 hospital deaths.

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